FREE-PISTON STIRLING ENGINES
AN ADVANCED POWER-CONVERSION SYSTEM FOR RESIDENTIAL AND COMMERCIAL BUILDINGS

Thomas J. Marusak
Mechanical Technology Incorporated

ABSTRACT

Stirling engines are currently under development for a variety of applications that require high efficiency and fuel indifference. The Free-Piston Stirling Engine (FPSE) is a uniquely configured power converter that operates on Stirling-cycle principles while incorporating only two moving parts for simplicity and extended life and reliability. The FPSE has proven qualities that ideally suit it for heat pumping and cogeneration application in buildings.

This paper addresses three separate and distinct aspects of the FPSE. Its physical characteristics and operating advantages are discussed first. Laboratory data is presented for a number of engines regarding life, efficiency, power, control, etc. The engine's development status is characterized next. Major research programs supporting FPSE development are outlined in terms of objectives, status, and time frame. Third, an economic assessment is presented for applications involving space conditioning of buildings. Economic projections are detailed for FPSE heat pumps and cogenerators showing the viability and limitations for each system.

INTRODUCTION

Stirling engines were first developed for practical use during the early 1800's, but lost their prominence with the advent of the lighter weight, less expensive internal-combustion engines. Today, however, Stirling engines are the focus of renewed interest and development activity. The renaissance of the Stirling engine is primarily due to the potential benefits stemming from two of the engine's distinguishing traits: alternative fuel capability and high efficiency.

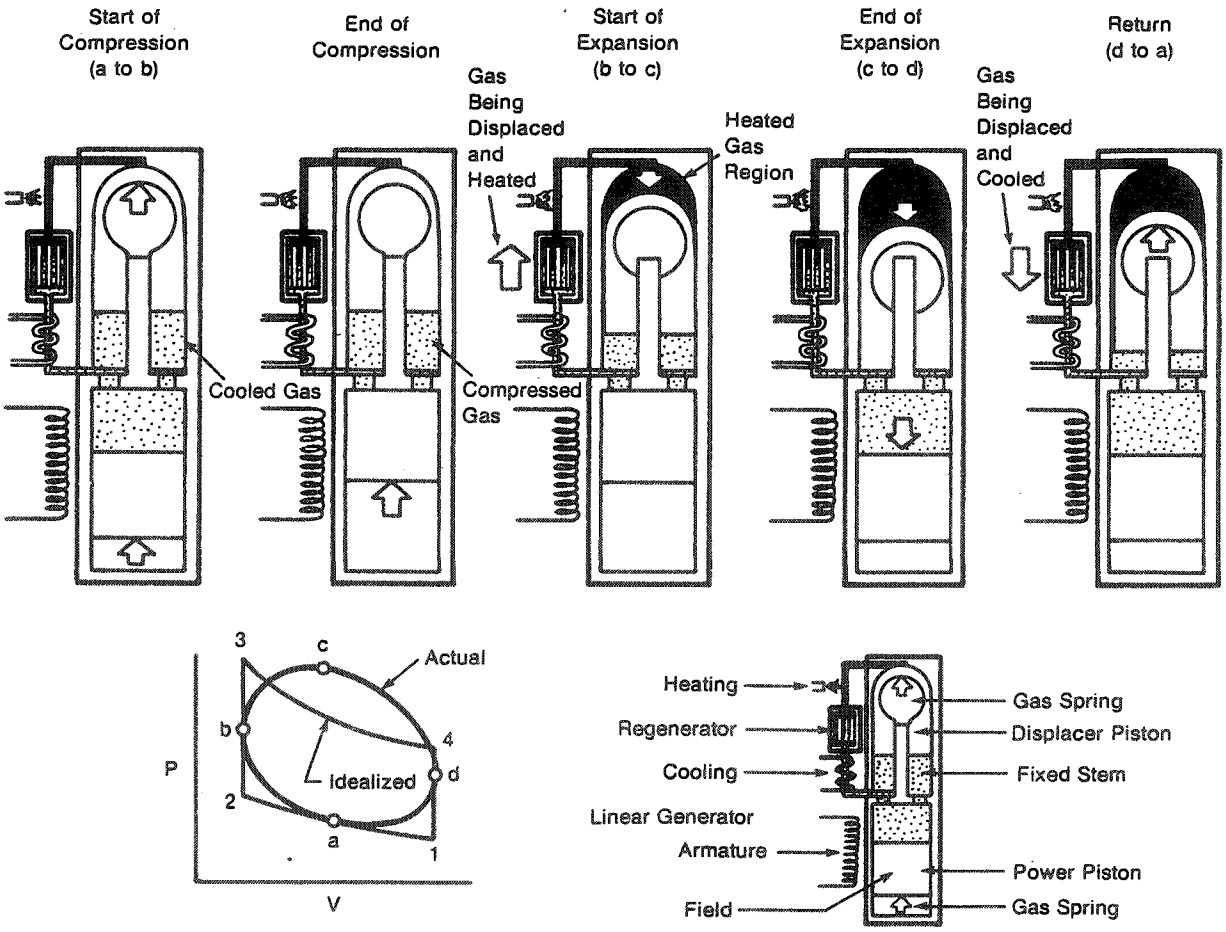
The Stirling engine's external heat input feature and Carnot cycle equivalency are the basis for the attributes of fuel indifference and high conversion efficiency. These two features make future applications possible with nonconventional energy sources such as solar, remote nuclear, solid fuels, agricultural fuels, and alike.

The FPSE is a modern, novel implementation of the Stirling cycle involving only two moving parts that can be floated on a film of working gas, virtually eliminating mechanical wear. Furthermore, oil lubrication is not required. Figure 1 illustrates the thermodynamic operating principles of the FPSE.

Unlike rotating machinery, the FPSE's dynamic parts move strictly in a reciprocating fashion. Further, the engine has no need for drive shaft seals in that its working gas (usually helium) is hermetically sealed within the engine at relatively high charge pressure (typically set at 1000 psia or more). The hermetic seal and reciprocating motion add to the engine's life and reliability characteristics, but they also impose difficult constraints on the integration of the engine with its load. Lacking a rotating drive shaft, the FPSE must be directly coupled to the loading device within the hermetic boundary of the engine and load combination.

Working within this constraint, designers have had to devise novel, innovative methods for transmitting the power generated by the thermodynamic cycle to the mechanical load. As a result, a variety of linear devices such as alternators, compressors, and pumps have been developed for integration with the FPSE. Figure 2 schematically illustrates a few of the various loading devices currently under development.

Due to the attractive advantages offered by the FPSE, i.e., multifuel capability, high efficiency and reliability, and long life, it is being developed and evaluated for a number of advanced power-conversion applications[1]. These application targets include advanced residential and commercial space-conditioning systems, cogenerators, military portable generators, and power systems for space stations and satellites.



The power piston/armature of this unusual stirling engine has no connecting rod, but generates electricity as it oscillates past the field windings. Simplified operation is shown here: (a) to (b)—gas spring below power piston/armature forces piston upward to generate electricity in the field windings and to compress the working gas between power piston and displacer piston; (b) to (d)—compressed gas, displaced through external passage, is heated to increase its volume, thus driving displacer piston and power piston/armature downward, generating electricity again; (d) to (a)—gas spring inside displacer piston forces piston upward to original position; working gas is displaced through the external passage and cooled in preparation for next compression cycle. Any waste heat can warm building, heat water, or be rejected.

Figure 1. FPSE operating principles.

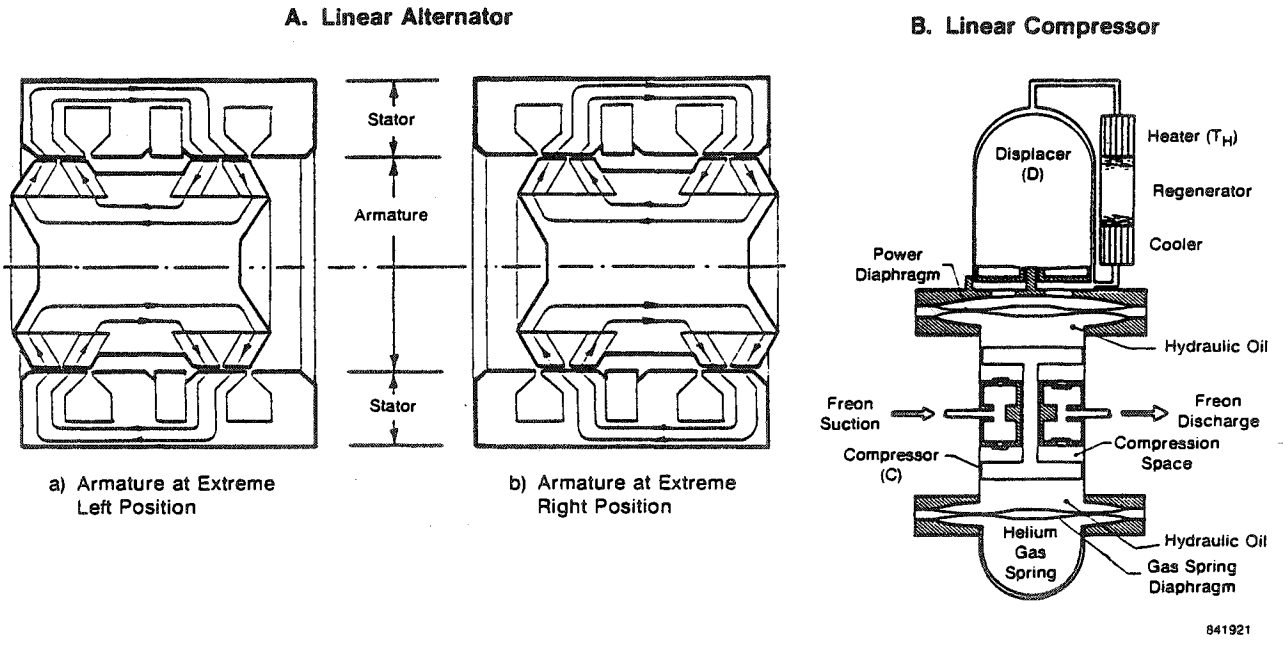


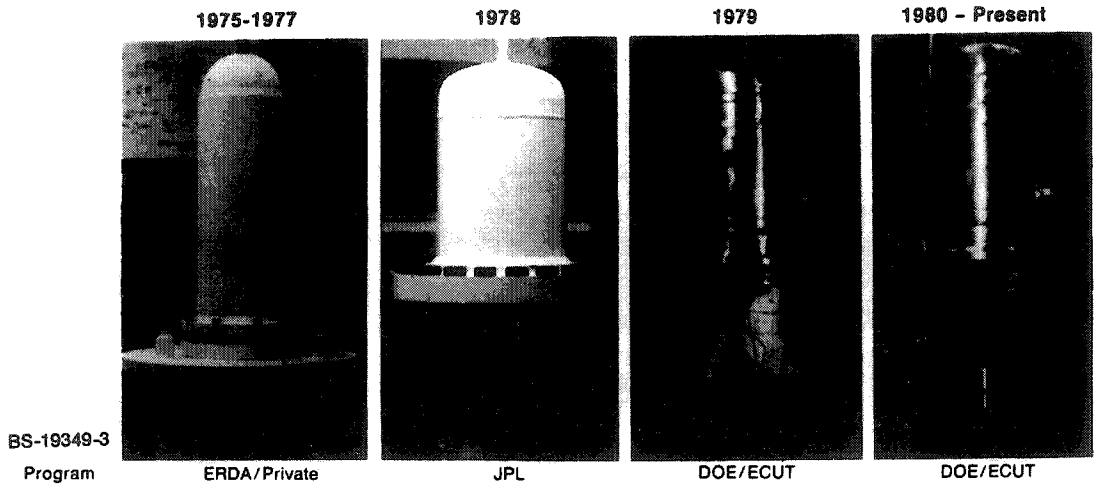
Figure 2. FPSE load configurations.

DEVELOPMENT STATUS

Nearly ten years ago FPSE feasibility demonstrators were being investigated for their practicality and application potential. These early units, found mainly in a few university engineering laboratories, were desk-top models with power capability below 10 watts[2]. Today, prototype systems are undergoing extensive development in power sizes ranging up to 25-kW electric power output.

Mechanical Technology Incorporated (MTI) employs the largest known research group in the world for FPSE development. Much of the critical technology research was completed at MTI between 1975 and 1980. Figure 3 illustrates some of the early-generation research engines that were developed during that time frame. Since then, MTI has entered into a stage of early product development utilizing a series of engineering prototype models. The four Engineering Model (EM) series engines have been coupled to both linear freon compressors and linear alternators. They have been operated with natural gas and diesel fuel, and one is currently being adapted for operation with solid fuel. Over 3000 hours of testing have been accumulated on the EM series, with EM #2 currently undergoing a 9000-hour continuous-operation test.

As shown in Figure 4, the efficiency level of the EM is competitive with alternative internal-combustion engines in its power class (4 hp).



	1975-1977	1978	1979	1980 - Present
Program	ERDA/Private	JPL	DOE/ECUT	DOE/ECUT
Purpose	Feasibility	Solar/Demonstration	Technology Research	Thermodynamic Research
Engine Power	0.5 to 1.0 kW	1.3 kW / 0.9 kW_e	1.7 kW / 1.1 kW_e	1.7 kW / 1.1 kW_e
Cycle Efficiency	0.22 to 0.27	0.30 to 0.34	0.39	0.39
Frequency	18 - 20 Hz	30 Hz	45 Hz	45 Hz
Load	Helium Damper	Alternator	Alternator	Alternator
Heat Input	Electric	Electric	Electric	Natural Gas

Figure 3. Evolution of FPSE technology research.

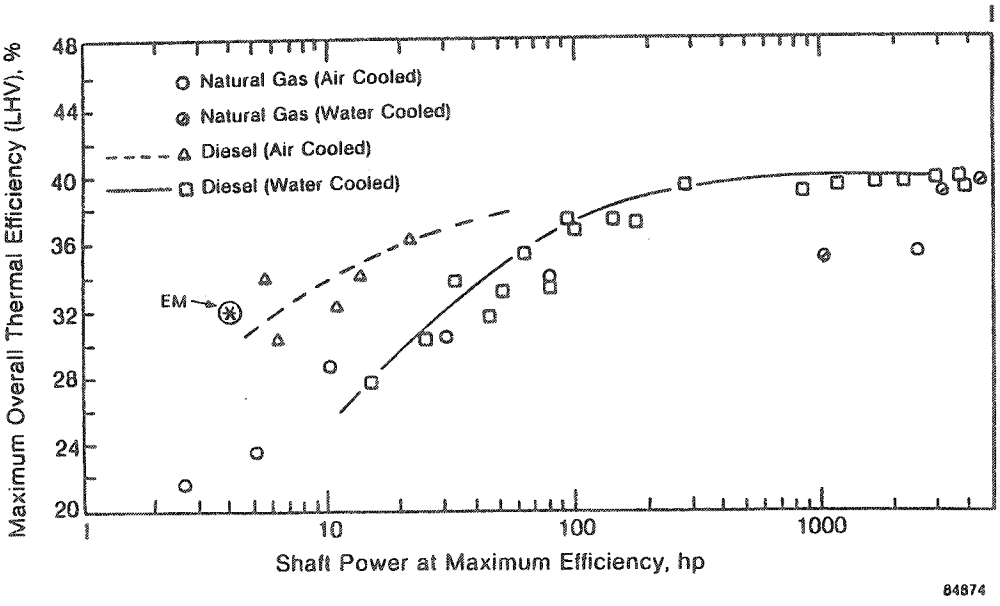


Figure 4. FPSE efficiency relative to diesel (and gas) engine overall thermal efficiency (LHV).

APPLICATIONS

Based on the successful development record of the EM series, aggressive application programs are now in progress. Three well-focused efforts are in place to develop gas-fueled heat pumps, military generators, and space-power systems. This section briefly describes some of the activities involved in

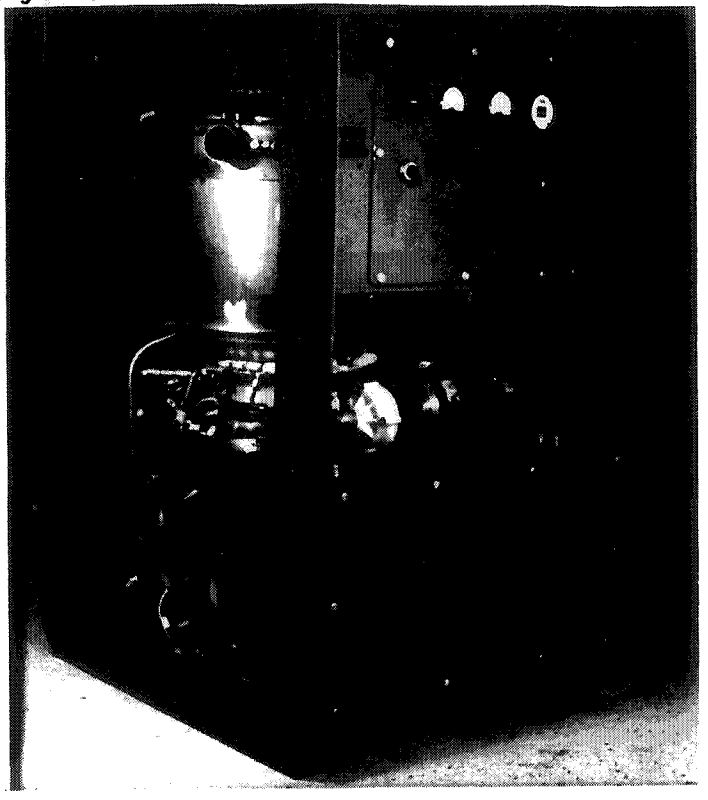
the latter two applications, and more thoroughly details the development program to evolve an advanced residential space-conditioning system.

Military Generator

Development of a reliable, multifuel, quiet engine-generator for tactical electric power is a continuing objective of the U.S. Army, Ft. Belvoir Research & Development Center. Since the 1960's, the Army has considered the Stirling-cycle engine as a means of achieving these objectives. To prove the military worth of the FPSE for this application, MTI was awarded a two-year contract to design, fabricate, develop, test, and deliver two 3-kW FPSE/linear alternator generator sets with the following characteristics:

- 3-kW net electric power at 0.8 power factor;
- military electrical transient requirements (voltage, frequency, waveform);
- diesel-fuel combustor;
- 250 hours mean time between failure;
- minimum 3000 hours of operation before overhaul; and,
- low noise (inaudible at 100 meters).

The military unit designated as the Advanced Development Model (ADM) incorporates design features which are evolutionary to that of the EM, extending even further the engine's reliability and life capability. Figure 5 is a picture of the ADM on test. This unit is scheduled to be delivered to the army for evaluation this year.



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Figure 5. ADM System on test.

Space-Power System

The continuing exploration of outer space presents an ever increasing need for reliable, long-life power sytems that can operate on nuclear or solar heat sources. As power requirements approach 100 kW or greater, the feasibility of photovoltaic power systems becomes less likely. As an investigation of the potential use of dynamic systems in space, NASA/Lewis Research Center has identified the FPSE as an attractive candidate for development. Addressing this need, MTI is developing a 25-kW electric generator for space application. As in the case of the military engine, the EM design features form the basis for the space-power system. The space engine involves two identical, opposed FPSE's providing dynamic balance and component symmetry. The first unit is scheduled to be tested during February, 1985.

Space-Conditioning Systems

Due to the FPSE's potential for long, reliable life, high efficiency, and clean, quiet operation, it is being developed for advanced space conditioning of buildings, with current emphasis on heat pumps for the residential sector[4]. Under cooperative funding from the Gas Research Institute and the U.S. Department of Energy[5], a program has been underway at MTI since January, 1983 to develop a prototype gas-fueled, residential energy-conversion system. During the early portion of the program, an analytical evaluation was conducted to determine the system configuration with the best opportunity for early product introduction. MTI's commercialization partner, Lennox Industries, lead in the product assessment.

For purposes of the study, the FPSE was considered generically as a gas-fueled prime mover that could be utilized in a number of system configurations. Three space-conditioning systems were evaluated in detail for economic potential. These systems included gas-fueled heat pumps, cogenerators, and hybrid heat pumps. The latter system involves the conversion of gas to electricity which, in turn, is used to operate an electric heat pump.

An energy-economics computer model was utilized to evaluate the economic performance of each FPSE system relative to two reference systems: 1) an advanced electric heat pump; and, 2) a pulse-combustion gas furnace and high-efficiency electric air conditioner. The model took into account system performance estimates obtained from prototype test data and design predictions. Figure 6a illustrates the range of engine efficiency used in the performance model relative to the measured EM effective mechanical efficiency data. The baseline refrigeration loop performance for the FPSE heat pump was determined from test data obtained from an FPSE prototype heat pump system under development at MTI. The data measured with the MTI/FPSE heat pump prototype is shown in Figure 6b.

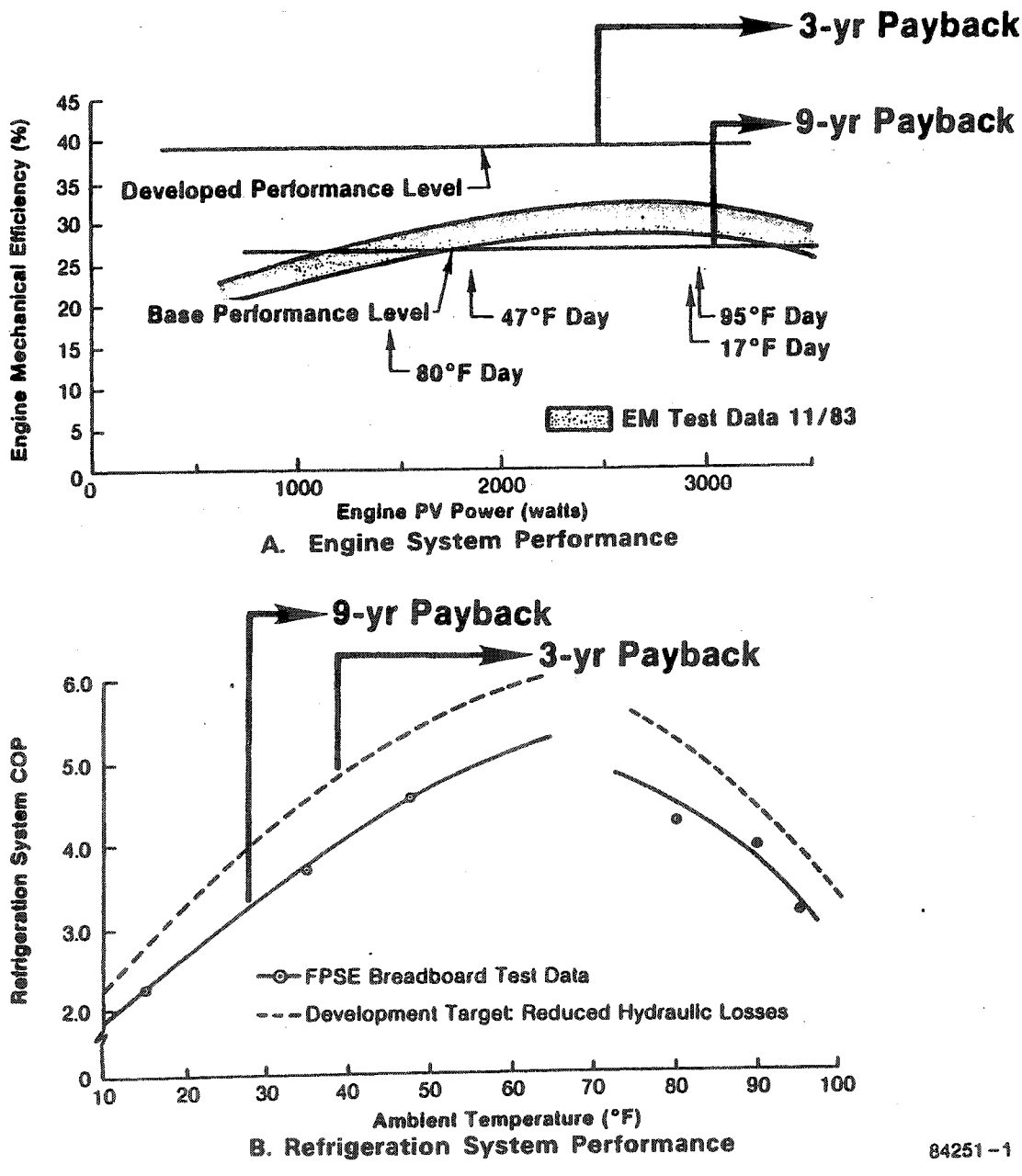
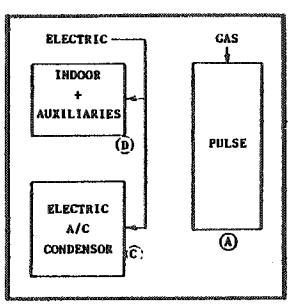


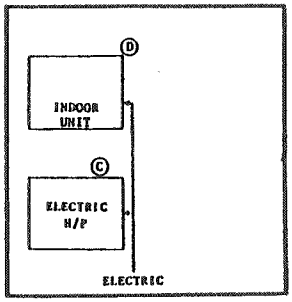
Figure 6. Performance status - product potential.

An economic analysis was performed to determine the annual energy costs for each of the three FPSE systems, as well as for two reference systems: a high-efficiency electric heat pump, and a pulse-furnace/electric air conditioner. Cost estimates for each of the FPSE systems were derived from manufacturing studies conducted by an independent engineering firm. The manufacturing costs were adjusted to reflect the consumer-level prices (not including their installation), which are shown in Table I.

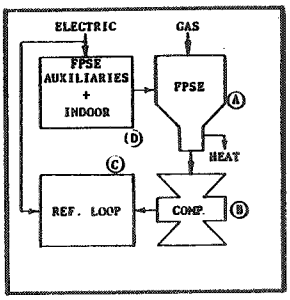
Table I. Comparison of system costs*.



Pulse & A/C



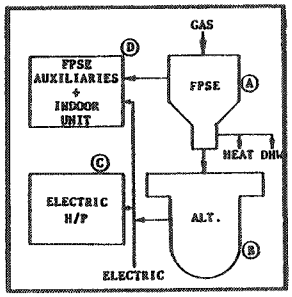
Electric H/P



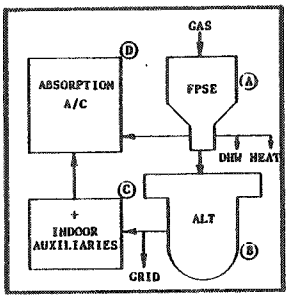
HAHP

Consumer-Level Costs - 1983 Dollars

Component			
A	1016	--	1362
B	--	--	919
C	1649	2129	997
D	265	362	572
Total	2920	2491	3850
Premium 1	Ref. 1	--	938
Premium 2	--	Ref. 2	1359



Hybrid H/P



Cogenerator

Consumer-Level Costs - 1983 Dollars

Component		
A	1362	1362
B	950	950
C	2129	4530
D	798	798
Total	5240	7640
Premium 1	2320	4720
Premium 2	2749	5149

*Consumer-level costs for equipment only (installation not included).

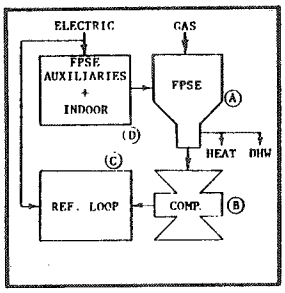
Energy prices used for this study were obtained via phone contact with the local utility serving each of the locations considered. Where applicable, life-line and demand rate schedules were taken into account. Although introduction of FPSE products is not anticipated until the 1989 time frame, escalation of energy prices was not considered, i.e., the results reflect the economic potential of the FPSE based on the 1983 energy situation, and with 1983 material/labor costs. The results of the study are displayed in Tables II and III. The projected annual energy input costs for each system are shown in Table II.

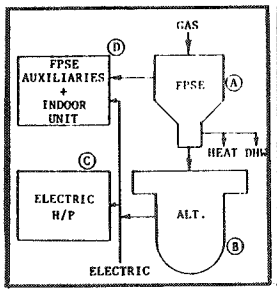
Table II. System input energy cost composite ($/year).

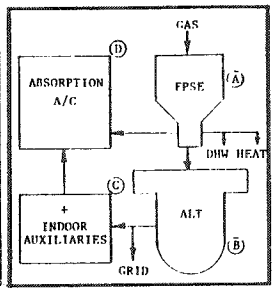
	Heat			Cool			
	Gas	Electric	Total	Gas	Electric	Total	Annual
Sunneyvale, Ca.							
Pulse and A/C	518	37	555	--	62	62	617
HAHP	206	65	271	35	17	52	323
Hybrid	404	--	404	78	1	79	483
Cogenerator	1541	-1195	346	222	-111	111	457
Electric H/P	--	695	695	--	144	144	839
Chicago							
Pulse and A/C	958	79	1037	--	220	220	1257
HAHP	586	231	817	97	72	169	986
Hybrid	837	5	842	265	18	283	1125
Cogenerator	1790	-1237	553	449	-263	186	739
Electric H/P	--	924	924	--	366	366	1290
Boston							
Pulse and A/C	1001	103	1114	--	164	164	1278
HAHP	636	122	758	95	46	140	898
Hybrid	839	5	842	191	5	196	1038
Cogenerator	1856	-1755	101	321	-231	90	194
Electric H/P	--	1837	1839	--	232	232	2069
New York							
Pulse and A/C	934	100	1034	--	188	188	1222
HAHP	562	275	837	69	51	120	957
Hybrid	780	5	785	182	4	186	971
Cogenerator	1848	-1761	87	302	-216	86	173
Electric H/P	--	1499	1499	--	282	282	1781
Dallas							
Pulse and A/C	358	29	387	--	356	356	743
HAHP	185	80	265	190	101	291	556
Hybrid	278	4	282	436	26	462	744
Cogenerator	725	-598	127	747	-417	330	457
Electric H/P	--	297	297	--	272	272	569

Table III is a compilation of the simple payback periods calculated for each system relative to the gas-fueled (pulse-furnace) reference system, and for the HAHP relative to the electric heat pump option.

Table III. System payback comparison for various locations.
Energy Prices: 12/83 - Performance: Best Projected







	HAHP Relative to Pulse & A/C	HAHP Relative to Elec.H/P	Hybrid Relative to Pulse & A/C	Cogenerator Relative to Pulse & A/C
Price Premium*	$938	$1359	$2320	$4720
Sunneyvale, Ca.				
Annual Energy Savings	$294	$ 516	$ 134	$ 160
Payback Period	3.1 Yrs	2.7 Yrs	17 Yrs	30 Yrs
Chicago				
Annual Energy Savings	$271	$ 304	$ 132	$ 518
Payback Period	3.4 Yrs	4.5 Yrs	17 Yrs	9 Yrs
Boston				
Annual Energy Savings	$380	$1171	$ 241	$1088
Payback Period	2.4 Yrs	1.2 Yrs	10 Yrs	4.3 Yrs
Long Island				
Annual Energy Savings	$265	$ 824	$ 251	$1049
Payback Period	3.5 Yrs	1.6 Yrs	9 Yrs	4.5 Yrs
Dallas				
Annual Energy Savings	$187	$ 13	$ 0	$ 236
Payback Period	5.0 Yrs	--	--	16 Yrs

*See Table I for derivation of price premium.

Based on these results, the following conclusions were established:

- out of the three FPSE system configurations considered, the gas-fueled heat pump is identified as showing the lowest calculated payback period with respect to the reference sytems;

- the gas-fueled heat pump is projected to be competitive with electric heat pumps in most areas of the country, except where the heating-to-cooling and electric-to-gas price ratios are low;
- the hybrid heat pump showed the highest calculated payback period, under no conditions reaching a value of five years or less; and,
- relative to the gas furnace and electric air conditioner baseline system, the heat/cool cogenerator showed marginally attractive payback (five years) only in the northeastern cities of New York and Boston where electricity is very expensive relative to gas (long payback periods were projected for areas where the electricity-to-gas price ratio is more moderate).

Due to the attractive payback period calculated for the gas-fueled heat pump system, it, rather than the hybrid or cogenerator, was selected as the prime system configuration for early product development. It should be pointed out that the technology development required for the cogenerator application is continuing under the Army's field generator program. As institutional barriers to small-scale cogeneration begin to fall, the FPSE could be introduced as a second-generation product following the gas-fueled heat pump.

In summary, the FPSE is a versatile, highly attractive prime mover with a broad-based application potential. The current development emphasis is on specialty-application, electric-generator and residential gas-fueled heat pumps. As the technical maturation process continues and larger power machines become available, the FPSE could become the cornerstone component for small-scale, advanced residential and commercial space-conditioning systems, including cogenerators and total energy systems.

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